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Semiconductor device and method for fabricating the same

Abstract

A semiconductor device includes a semiconductor substrate, a circuit structure and a ring-shaped protrusion. The semiconductor substrate has a front surface and a rear surface opposed to each other. The circuit structure is located on the front surface. The ring-shaped protrusion is protruded on the rear surface.

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Background/Summary

(1) This application claims the benefit of People's Republic of China application Serial No. 202011319465.5, filed Nov. 23, 2020, the subject matter of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The disclosure relates in generally related to a semiconductor device and method for fabricating

the same, and more particularly to a semiconductor device having a ring-shaped protrusion and method for fabricating the same.

(3) In the manufacture of semiconductor devices, wafers need to be cut to separate into a plurality of individual dies. However, the stress caused by cutting may cause chipping and warping of the separated dies, and the chipped or warped dies may also be further warped due to the influence of the subsequent bonding wire connection or the shrinkage stress of the packaging material, resulting in a poor subsequent process yield.

(4) Therefore, there is a need of providing an improved semiconductor device and method for fabricating the same to obviate the drawbacks encountered from the prior art.

SUMMARY

(5) One aspect of the present disclosure is to provide a semiconductor device, wherein the semiconductor device includes a semiconductor substrate, a circuit structure and a ring-shaped protrusion. The semiconductor substrate has a front surface and a rear surface opposed to each other. The circuit structure is located on the front surface. The ring-shaped protrusion is protruded on the rear surface.

(6) Another aspect of the present disclosure is to provide a method for fabricating a semiconductor device, wherein the method comprises steps as follows: Firstly, a semiconductor substrate having a front surface and a rear surface opposite to each other is provided. A circuit structure is then formed on the front surface. Subsequently, a cutting process is performed to form at least one front recess on the front surface. An etching process is then performed to form at least one ring-shaped protrusion on the rear surface.

(7) By forming a ring-shaped protrusion on the rear surface, the stress resistance of the semiconductor substrate can be improved which can prevent the semiconductor substrate and the circuit pattern located on the front surface of the semiconductor substrate from being warped due to the adverse impacts caused by the subsequent process, such as bonding stress generated by a wire bonding process or shrinkage stress of the packaging material resulted by a packing process. Whereby a flatter process surface can be provided for the subsequent process and the yield thereof can be thus improved. In addition, plasma etching technology is used to replace the diamond cutter to cut the semiconductor substrate, the problem of chip collapse due to stress damage caused by cutting can be thus improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above objects and advantages of the present disclosure will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

(2) FIGS. 1A to 1F are schematic diagrams of a series of process structures for fabricating a semiconductor device in accordance with one embodiment of the present disclosure;

(3) FIG. 1D' is a top view illustrating an intermediated process structure of the semiconductor device depicted in FIG. 1D;

(4) FIG. 1E' is a top view illustrating an partial process structure of the semiconductor device depicted in FIG. 1E;

(5) FIG. 1F' is a top view illustrating an intermediate process structure of the semiconductor device depicted in FIG. 1F;

(6) FIG. 2 is a top view of a semiconductor device in accordance with another embodiment of the present disclosure;

(7) FIG. 3 is a top view of a semiconductor device in accordance with yet another embodiment of the present disclosure;

- (8) FIG. 4 is a top view of a semiconductor device in accordance with further embodiment of the present disclosure;
- (9) FIG. 5 is a top view of a semiconductor device in accordance with one embodiment of the present disclosure;
- (10) FIG. 6 is a top view of a semiconductor device in accordance with one embodiment of the present disclosure; and
- (11) FIG. 7 is a top view of a semiconductor device in accordance with one embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) The embodiments as illustrated below provide a semiconductor device with a ring-shaped protrusion and a method of fabricating the same. The ring-shaped protrusion can provide anti-bending properties to increase the stress resistance of the semiconductor substrate and prevent the semiconductor substrate itself and the circuit pattern located on the semiconductor substrate from being warped due to the adverse impacts caused by the subsequent process, such as bonding stress generated by a wire bonding process or shrinkage stress of the packaging material resulted by a packing process. Whereby, a flatter process surface can be provided for the subsequent process, and the yield thereof can be thus improved. In order to more clearly describe the above and other goals, features and advantages of this disclosure, the following describes the method of fabricating a semiconductor device with ring-shaped protrusions, as a preferred embodiment, and with the accompanying drawings, the detailed description is as follows.

(13) It is to be noted that the following descriptions of preferred embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed. Also, it is also important to point out that there may be other features, elements, steps and parameters for implementing the embodiments of the present disclosure which are not specifically illustrated. Thus, the specification and the drawings are to be regarded as an illustrative sense rather than a restrictive sense. Various modifications and similar arrangements may be provided by the persons skilled in the art within the spirit and scope of the present disclosure. In addition, the illustrations may not be necessarily drawn to scale, and the identical elements of the embodiments are designated with the same reference numerals.

(14) Referring to FIGS. 1A to 1F, FIGS. 1A to 1F are schematic diagrams of a series of process structures for fabricating a semiconductor device **100** in accordance with one embodiment of the present disclosure. In one embodiment of the present disclosure, as shown in FIG. 1A, a semiconductor substrate **101** having a front surface **101a** and a rear surface **101b** opposite to each other is firstly provided. A circuit structure **102** is then formed on the front surface **101a** of the semiconductor substrate **101**. A first adhesive layer **107** is formed on the rear surface **101b** of the semiconductor substrate **101**.

(15) In one embodiment of the present disclosure, the circuit structure **102** may be a patterned circuit layer formed on the front surface **101a** of the semiconductor substrate **101**. For example, in the present embodiment, the circuit structure **102** at least includes a plurality of circuit pattern units **102a** with the same circuit layout and arranged adjacently on the front surface **101a** of the semiconductor substrate **101**.

(16) Next, a cutting process is performed. Referring to FIGS. 1B and 1C, in one embodiment of the present disclosure, the cutting process includes forming at least one front recess on the front surface **101a**. In the present embodiment, the step of performing the cutting process includes performing a laser scribing **104** on the front surface to form a first sub-front recess **104a** (as shown in FIG. 1B), and then performing a mechanical dicing **105** to form a second sub-front recess **105a** (as shown in FIG. 1C), wherein the second sub-front recess **105a** is located at the bottom **104b** of the first sub-front recess **104a**; and the combination of the first sub-front recess **104a** and the second sub-front recess **105a** forms a front recess **110** of the front surface **101a**.

(17) It is noted that the order and number of the laser scribing **104** and the mechanical dicing **105**

for forming the front recess **110** and the cut size of the first sub-front recess **104a** and the second sub-front recess **105a** are not particularly limited. As long as the sub-front recesses (such as the first sub-front recess **104a** and the second sub-front recess **105a**) are aligned or overlapped with each other, those process used for forming the same do not depart from the spirit of the cutting process described in the present disclosure.

(18) Referring to FIG. **1D**, in one embodiment of the present disclosure, the first adhesive layer **107** is peeled from the rear surface **101b** of the semiconductor substrate **101** after the above-mentioned cutting process is carried out. Subsequently, in one embodiment of the present disclosure, the semiconductor substrate **101** is turned upside down so that the front surface **101a** faces downward and the rear surface **101b** faces upward, and then a second adhesive layer **108** is formed on the front surface **101a** of the semiconductor substrate **101**. At least one ring-shaped photoresist **109** is formed on the rear surface **101b** of the substrate **101**. The number of the ring-shaped photoresist **109** may be one, two, three, four or more, for example. For clarity of description, in the present embodiment, as shown in FIG. **1D** and FIG. **1D'** (top view of FIG. **1D**), merely two ring-shaped photoresists **109** are depicted. These two ring-shaped photoresists **109** are respectively located at two opposite sides of the front recesses **110** (such as, the combination of the first sub-front recess **104a** and both sides of the second sub-front recess **105a**) and are respectively at least partially overlapped with one of the two adjacent circuit pattern units **102a**. In another embodiment, only one ring-shaped photoresist **109** can be formed on either side of the front recess **110** (consisting of the first sub-front recess **104a** and the second sub-front recess **105a**) on the rear surface **101b**; and no any other ring-shaped photoresist **109** is formed on the other side of the front recess **110**.

(19) Then, an etching process **106** is performed. Referring to FIG. **1E**, at least one ring-shaped protrusion **103** is formed on the rear surface **101b**. In the present embodiment, an etching process **106** is performed to remove a portion of the semiconductor substrate **101** not covered by the ring-shaped photoresist **109** from the rear surface **101b**, so that the remaining portion of the semiconductor substrate **101** covered by the ring-shaped photoresist **109** forms a ring-shaped protrusion **103** relative to the etched rear surface **101b**. In addition, the etching process **106** further includes forming at least one rear recess **106a** in the etched rear surface **101b** to make the rear recess **106a** adjacent to the ring-shaped protrusion **103** and communicating with the front recess **110**. As a result, the semiconductor substrate **101** can be divided into a first substrate **120** and a second substrate **130** which are both adhered to the second adhesive layer **108**.

(20) After the etching process **106** is performed, the ring-shaped photoresist **109** is peeled off and the second adhesive layer **108** is removed from the front surface **101a**, the first substrate **120** and the second substrate **130** can be separated from each other. In the present embodiment, referring to FIG. **1F** and FIG. **1F'** (top view of FIG. **1F**), each of the first substrate **120** and the second substrate **130** separated from each other has a ring-shaped protrusion **103** formed by the remaining semiconductor substrate **101** on the rear surface **101b**.

(21) Specifically, the front surface **101a** of each of the first substrate **120** and the second substrate **130** includes a circuit pattern unit **102a** with the same circuit layout. The first substrate **120** and the second substrate **130** separated from each other can be regarded as two divided dies. Taking the first substrate **120** as an example of a semiconductor device, it includes a semiconductor substrate **101**, a circuit structure **102** and a ring-shaped protrusion **103**. The semiconductor substrate **101** has a front surface **101a** and a rear surface **101b** opposite to each other. The circuit structure **102** is located on the front surface **101a**. The ring-shaped protrusion **103** is protruded on the rear surface **101b**; wherein a distance PI from a top surface **103t** of the ring-shaped protrusion **103** to the front surface **101a** is greater than a thickness T of the semiconductor substrate **101** (also referred to as the height of the vertical wall **101c**) measured from the front surface **101a** to the rear surface **101b**.

(22) Subsequently, after series of downstream processes (not shown) are performed on the first substrate **120** and the second substrate **130**, the preparation of the semiconductor device **100** as shown in FIG. **1F** is completed.

(23) In some embodiments of the present disclosure, a thickness T of the semiconductor substrate **101** may be, for example, 100 micrometers, and a height H of the ring-shaped protrusion **103** may be, for example, between 20 micrometers and 50 micrometers. A ratio of the thickness T of the semiconductor substrate **101** to the height H of the ring-shaped protrusion **103** is substantially between 10:1 and 5:1, and may be, for example, 10:1, 9:1, 8:1, 7:1, 6:1 or 5:1. A region R surrounded by the ring-shaped protrusion **103** at least partially overlaps the circuit structure **102**. In one embodiment, the ring-shaped protrusion **103** can be arranged in adjacent to a vertical wall **101c** of the semiconductor substrate **101**, and the vertical wall **101c** connects the front surface **101a** to the rear surface **101b**, and an outer outmost edge **103o** of the ring-shaped protrusion **103** away from the center **103c** is separated from the vertical wall **101c** for a distance D (or a distance D' in the opposite side). In one embodiment, the front surface **101a** has a front recess **110**, and a bottom **104b** of the front recess **110** is connected with the vertical wall **101c**, so that the bottom **104b** of the front recess **110**, the vertical wall **101c**, and a part of the rear surface **101b** connected with the vertical wall **101c** jointly define one protrusion **101p**.

(24) It is noted that the structure of the ring-shaped protrusion is not limited to this regard. For example, FIG. 2 is a top view of a semiconductor device **200** in accordance with another embodiment of the present disclosure. The structure of the semiconductor device **200** is generally similar to the semiconductor device **100** shown in FIG. 1F', except that in the present embodiment, the ring-shaped protrusion **203** of the semiconductor device **200** further includes a first extending protrusion **203a** extending from an inner edge **203s** of the ring-shaped protrusion **203** towards a center **203c** of the ring-shaped protrusion **203**.

(25) Referring to FIG. 3, FIG. 3 is a top view of a semiconductor device **300** in accordance with another embodiment of the present disclosure. The structure of the semiconductor device **300** is generally similar to the semiconductor device **200** shown in FIG. 2 except that in the present embodiment, two ends of the first extending protrusion **303a** of the ring-shaped protrusion **303** of the semiconductor device **300** are both connected to the inner edge **103s** of the ring-shaped protrusion **303**, and a region R3 surrounded by the ring-shaped protrusion **303** is divided into two sub-regions R31, R32.

(26) Referring to FIG. 4, FIG. 4 is a top view of a semiconductor device **400** in accordance with further embodiment of the present disclosure. The structure of the semiconductor device **400** is generally similar to the semiconductor device **300** shown in FIG. 3, except that in this embodiment, the ring-shaped protrusion **403** of the semiconductor device **400** further includes a second extending protrusion **403b**, wherein the first extending protrusion **403a** and the second extending protrusion **403b** come cross with each other to form a grid structure dividing the region R4 surrounded by the ring-shaped protrusion **403** into a plurality of sub-regions (such as sub-regions R41, R42, R43, R44).

(27) Referring to FIG. 5, FIG. 5 is a top view of a semiconductor device **500** in accordance with still further embodiment of the present disclosure. The structure of the semiconductor device **500** is generally similar to the semiconductor device **400** shown in FIG. 4, except that in the present embodiment, the ring-shaped protrusion **503** of the semiconductor device **500** further includes more extending protrusions, such as lateral extending protrusions **503a** and longitudinal extending protrusions **503b**, **503c**, and **503d**. Wherein, the lateral extending protrusion **503a** and the longitudinal extending protrusions **503b**, **503c**, **503d** come cross with each other to form a grid structure dividing the region R5 surrounded by the ring-shaped protrusion **503** into a plurality of sub-regions (such as sub-regions R51, R52, R53, R54, R55, R56, R57, R58).

(28) Although the shape of the ring-shaped protrusion **103** shown in FIGS. 2 to 5 is square, the shape of the ring-shaped protrusion is not limited to these regards. For example, referring to FIG. 6, the structure of the semiconductor device **600** is generally similar to the semiconductor device **100** depicted in FIG. 1F', except that in the present embodiment, the shape of the ring-shaped protrusion **603** of the semiconductor device **600** can be, for example, a circle.

(29) Referring to FIG. 7, the structure of the semiconductor device **700** is generally similar to the semiconductor device **600** shown in FIG. 6, except that in the present embodiment, the ring-shaped protrusion **703** of the semiconductor device **700** includes first extending protrusion **703a** and second extending protrusion **703b** coming cross with each other to divide the region R7 surrounded by the ring-shaped protrusion **703** into a plurality of sub-regions (for example, the sub-regions R71, R72, R73, R74).

(30) It is to be expressly understood that such is only an example of the shape, size and distribution of the ring-shaped protrusions of the semiconductor devices. The ring-shaped protrusions may be of dissimilar shape and/or dimensions. For example, some may be quadrilateral in shape but some may have sides of the same dimension and some may have sides of different dimensions. Included within such shapes are squares, rectangles, parallelogram, rhombus, trapezoids and kite.

Alternatively, the ring-shaped protrusions may be defined by a number of sides other than four, such as triangles, pentagons, hexagon, heptagons, and octagons, etc. The ring-shaped protrusions may also be of curved configuration, including circular, elliptical, arched, lens, crescent, oval, quatrefoil, and other curved shapes. Similarly, the shape of the region or sub-region surrounded by the ring-shaped protrusion can also be the above-mentioned shape, which will not be repeated here.

(31) In the process of fabricating the semiconductor devices **200** and **300** as shown in FIGS. 2 to 3, the ring-shaped photoresist **109**, used to form the ring-shaped protrusions **203** and **303**, includes a ring-shaped portion and an extending portion. The extending portion extends toward inside the ring-shaped portion.

(32) In the process of fabricating the semiconductor devices **400**, **500**, and **700** as shown in FIGS. 4 to 5 and 7, the ring-shaped photoresist **109**, used to form the ring-shaped protrusions **403**, **503**, and **703**, includes a ring-shaped portion and a grid portion, the grid portion is located inside the ring-shaped portion and connected to the ring-shaped portion. As such, the plurality of extending protrusions of the ring-shaped protrusions **403**, **503**, and **703** could come cross with each other to form a grid structure, and the regions R4, R5, and R7 surrounded by the ring-shaped protrusions **403**, **503**, and **703** can be divided into a plurality of sub-regions by the grid structure.

(33) Based on the above, in the embodiments of the present disclosure, plasma etching technology is applied to replace the traditional way of relying a cutter (such as a diamond cutter) to cut the semiconductor substrate, whereby the problem of chip collapse due to the stress damage caused by the cutter cutting can be avoided. In addition, by forming a ring-shaped protrusion on the rear surface, the stress resistance of the semiconductor substrate can be improved, which can prevent the semiconductor substrate and the circuit pattern located on the front surface of the semiconductor substrate from being warped due to the adverse impacts caused by the subsequent process, such as bonding stress generated by a wire bonding process or shrinkage stress of the packaging material resulted from a packing process. Whereby a flatter process surface can be provided for the subsequent process and the yield thereof can be thus improved. The semiconductor-based components with ring-shaped protrusions described in the embodiments of this disclosure can be applied to flip-chip products (such as underfill and heatsink) and wire bonding products (such as Epoxy Resin type) and Die Attached Film type (DAF type).

(34) While the disclosure has been described by way of example and in terms of the exemplary embodiment(s), it is to be understood that the disclosure is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

Claims

1. A semiconductor device, comprising: a semiconductor substrate, having a front surface and a rear surface opposed to each other, wherein a vertical wall of the semiconductor substrate connects

the front surface to the rear surface; the front surface has a front recess; and a bottom of the front recess connects to the vertical wall; a circuit structure, located on the front surface; and a ring-shaped protrusion, protruding on the rear surface and having a height between 20 micrometers and 50 micrometers, wherein an outer outmost edge of the ring-shaped protrusion is separated from the vertical wall for a first distance; and a second distance between a top surface of the ring-shaped protrusion to the front surface is greater than a height of the vertical wall measured from the front surface to the rear surface.

2. The semiconductor device according to claim 1, wherein a ratio of a thickness of the semiconductor substrate to a height of the ring-shaped protrusion is between 10:1 and 5:1.
 3. The semiconductor device according to claim 1, wherein an area surrounded by the ring-shaped protrusion at least partially overlaps the circuit structure.
 4. The semiconductor device according to claim 1, wherein a shape of the ring-shaped protrusion is square or circular.
 5. The semiconductor device according to claim 1, wherein the ring-shaped protrusion further comprises a first extending protrusion extending from an inner edge of the ring-shaped protrusion towards a center of the ring-shaped protrusion.
 6. The semiconductor device according to claim 5, wherein two ends of the first extending protrusion are both connected to the inner edge, and a region surrounded by the ring-shaped protrusion is divided into two sub-regions.
 7. The semiconductor device according to claim 6, wherein the ring-shaped protrusion further comprises a second extending protrusion, and the region is divided into a plurality of sub-regions by the first extending protrusion and the second extending protrusion.
 8. The semiconductor device according to claim 1, wherein the ring-shaped protrusion is arranged adjacent to the vertical wall of the semiconductor substrate.
 9. The semiconductor device according to claim 8, wherein the bottom of the front recess, the vertical wall and a portion of the rear surface jointly define a protrusion.
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